Answer all the questions in this section in the spaces provided.

1 **(a)** Define moment of a force.

.....

..... [1]

(b) Fig. 1.1 shows a uniform metal beam pivoted at one end O and held in equilibrium in a horizontal plane by a spring attached to its other end P.

The spring has an unstretched length of 7.0 m and obeys Hooke's law.

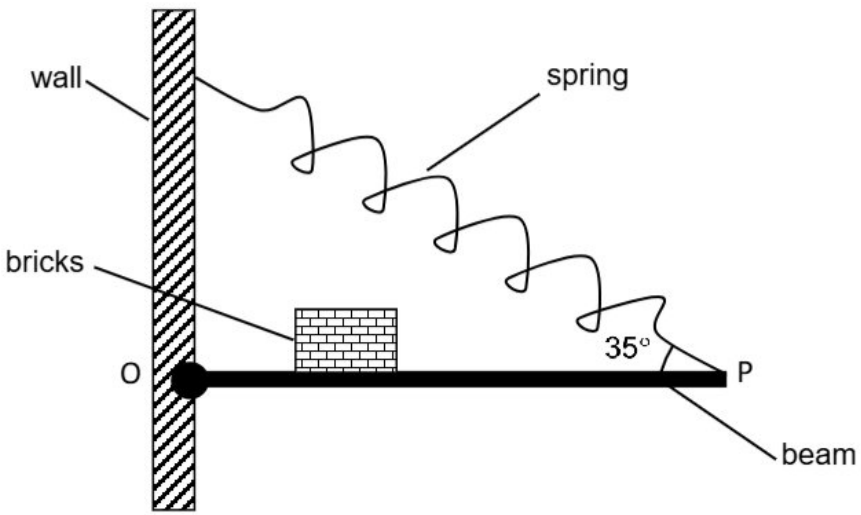


Fig. 1.1 (not to scale)

The beam has mass 190 kg and length 6.8 m. The spring makes an angle of 35° to the beam. A pile of bricks of total mass 120 kg is placed on the beam.

(i) Show that the extension of the spring is 1.3 m. [1]

(ii) On Fig. 1.1, draw labelled arrows to show all the forces acting on the beam. [2]

(iii) Right now, the centre of the pile of bricks is 5.0 m away from end P. Calculate the spring constant k of the spring.

k = ... N m^{-1} [2]

(iv) Calculate the elastic potential energy E_P of the spring.

E_P = ... J [2]

(v) The spring in Fig. 1.1 is now replaced with an inextensible string such that the beam remains in the original horizontal position.

Describe and explain what would happen to the force exerted by the string on the beam as the pile of bricks is slowly moved towards the pivot O while maintaining equilibrium.

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..... [2]

[Total: 10]